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Artificial Intelligence Integration in English Reading Pedagogy: A Systematic Review (2019-2024)

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ABSTRACT

In the education sector, numerous research studies have testified the effectiveness of using artificial intelligence (AI) to enhance English language skills. Still, there are very few review studies that specifically focus on the role of AI in teaching and learning English reading skills. Therefore, this review attempts to fill this gap and conducts a systematic review of the articles published between 2019 and 2024 on the use of AI in English reading (AIER). This study identifies the integration and efficacy of AI applications in reading pedagogy; and presents direction for future exploration. The researchers followed article selection, article screening & inclusion and data coding & analysis steps according to the PRISMA approach for the review process. The analysis indicates that a variety of AI-supported tools and platforms are used by both educators and students to support reading instruction. These tools and platforms are mostly used to create adaptive reading assignments, offer tailored feedback, and enhance access to educational materials. Instead of taking the place of instructional decision-making in English reading pedagogy, these tools are frequently employed to promote learner autonomy and engagement. Moreover, this review also presents the issues that should be addressed to obtain the utmost result of using AI in this field. Hence, the review analysis suggests that AI, being used effectively in teaching English reading skills across the globe, has transformational prospects for institutions and educators of ESL or EFL countries to utilize the scope of employing the AI teaching tactics in improving English reading pedagogy.

Key words: Artificial Intelligence (AI), Review, English Reading Pedagogy, Systematic Review

INTRODUCTION

Advanced technologies in education have been changing the dynamics of teaching and learning. The growing AI technology, and its applications in education, is leading to prospective futures in terms of personalized learning, machine dependent pedagogical systems, dynamic evaluation, meaningful interaction facilitation and so on, in online or hybrid digitized learning environments Although AI in education has been thoroughly examined, current syntheses mainly focus on STEM fields or general language proficiency, providing little understanding of how AI particularly affects cognitive, pedagogical, and assessment processes in English reading. Furthermore, there is a conceptual gap that this study aims to fill because few evaluations incorporate these applications into accepted theories of second language reading development.

Reading in English is one of the crucial skills for gaining English proficiency. Reading proficiency is fundamental for acquiring knowledge and for better academic achievement (Mahfouz & Mohamed, 2023). In teaching English reading skill, pedagogical effectiveness has gained a lot of attention since it significantly and consistently raises students' literacy levels, which in turn, promotes excellence in language skills (Gallagher et al., 2016). In recent years, AI in ELT research, especially in teaching English reading skills, has shown promising results with novel applications in improving speech recognition, translation, learning English language skills, assessment, learners' satisfaction and attitude and so on. For example, Hidayat (2024) highlights in his research that the implementation of AI platforms based on AI technology can customize learning for every student, detect specific student difficulties, and offer prompt feedback, all of which enhances reading comprehension. By artificial intelligence-powered speech recognition technology, Bachiri et al. (2024) revealed that AI-enabled speech recognition assessments demonstrated enhancement in foundational literary development, significant reading achievement, and improvement in readers' engagement with instant feedback with under-resourced education system. These results demonstrate the prospects of using AI technology in improving English reading skills and a promising potential for

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research on AI applications in reading pedagogy. However, there are a few review studies that specifically addressed AI in English reading pedagogy. The literature review studies that are published between 2019 and 2024 focusing on AI in English reading teaching and learning mostly present machine reading comprehension (RC) datasets produced by the NLP community and identified key dimensions in terms of format, domain, internal architecture, challenges, providing a conceptual framework for evaluating current and future resources (Rogers et al., 2023; Baradaran et al., 2022; Liu et al., 2019). Although Zhao and Nazir (2022) highlight the role of AI in improving learners' reading efficiency, prior reviews have tended to concentrate on individual results rather than providing a systematic, domain-based study of AI applications in English reading instruction. By combining research from the realms of learning, education, and assessment, this review addresses that constraint.

This systematic review uses a theoretical framework that brings together sociocultural theory, self-regulated learning theory, and the Technological Pedagogical Content Knowledge (TPACK) model. Sociocultural theory posits that learning develops through interaction and support and in this sense AI tools can be seen as valuable resources that assist learners by providing feedback and opportunities for reading practice (Vygotsky, 1978). Furthermore, self-regulated learning theory (Zimmerman, 2002) helps to explain how Artificial Intelligence may support learners through personalized feedback and making learners more engaged with materials. Alongside this, the TPACK framework highlights the central role of teachers in integrating AI meaningfully into classroom pedagogy. According to this model, technology should support pedagogical goals rather than replace teachers (Mishra & Koehler, 2006). Based on this combined framework, the review synthesizes empirical studies published between 2019 and 2024 and organizes the findings into three interconnected domains: learning, pedagogy, and assessment.

As few past studies have generally looked at the pedagogical elements of enhancing reading skills with the help of AI, this literature review will provide practitioners with an overview of AI technologies adopted in English reading pedagogy. This review also intends to present the efficacy of using AI technology in English reading pedagogy that is found in the primary research studies in the past six years and the scope of future educational research and practices in teaching and learning English reading skills in order to design more effective ELT classroom systems. This review will help educators to design pedagogy on reading skills, and will also enlighten different users about learning the overall usability of AI in improving reading skills. Hence, this review study, by using a systematic review model (Page et al., 2021), provides a synthesis of knowledge of the field from which future research priorities, directions can be identified. Therefore, the present study addresses the research gap with the help of the following research questions:

1. What aspects of English reading skills have been explored integrating Artificial Intelligence (AI) between 2019 and 2024?
2. What are the scopes for future research and practices in Artificial Intelligence integrated teaching and learning of English Reading skills?

METHODOLOGY

The researchers adopted the PRISMA (Preferred Reporting Items for Systematic reviews and Meta-Analyses) approach (Page et al., 2021) for article selection and analysis for this review. The review study included a total of 44 articles which focused on AI application in teaching and learning English reading skills published from 2019 to 2024. The review process followed article selection, article screening and inclusion and data coding and analysis steps.

Article Selection

As the objective of this paper is to find out the effectiveness of AI in improving English reading skills, journal articles published in this field based on primary research have been selected. The three researchers collaboratively set the criteria mentioned in Figure 1 for selecting the articles. The key words such as 'Artificial intelligence and English reading skills', 'Teaching English reading skills with AI', 'English Language Teaching with AI', 'Artificial Intelligence and English Language Teaching' were used for initial search. Articles were retrieved from the database 'Google Scholar' and 'Scopus'. The total number of articles in Google Scholar was 34, 900 (Artificial intelligence and English reading skills: 17,700 & English Language Teaching with AI: 17,200) and the total number of articles in Scopus was 667. The identification was customized according to time span 2018-2024, peer reviewed journal articles written in English. This exclusion resulted in 800 articles (600 Google Scholar articles and 200 Scopus articles) which were screened in the next phase.

Article Screening and Inclusion

In the second phase, researchers excluded articles which are not related to the use of AI in teaching and learning of English reading skills. The researchers also excluded secondary research papers, book chapters, commentary or theoretical articles, proceeding papers or thesis papers, which yielded 400 papers in total. All the papers were filed in a Google drive. Then, three researchers manually moved the duplicate articles into the trash and finalized 400 articles for final assessment. The researchers then read the abstracts, discussed among themselves, and sorted 210 primary research articles that employed AI intervention in improving English reading skills. A total of 125 full text articles were selected from Scopus publications as Scopus-indexed publications are generally considered as high-quality publications. Peer-reviewed articles that are selected from Google Scholar, but not Scopus-indexed, also have been reviewed as these articles presented primary research on the use of AI in improving reading skills, which directly matches with the theme of the current review. Finally, 44 primary research articles were analyzed for review on the use of AI in improving reading skills in English (Appendix A).

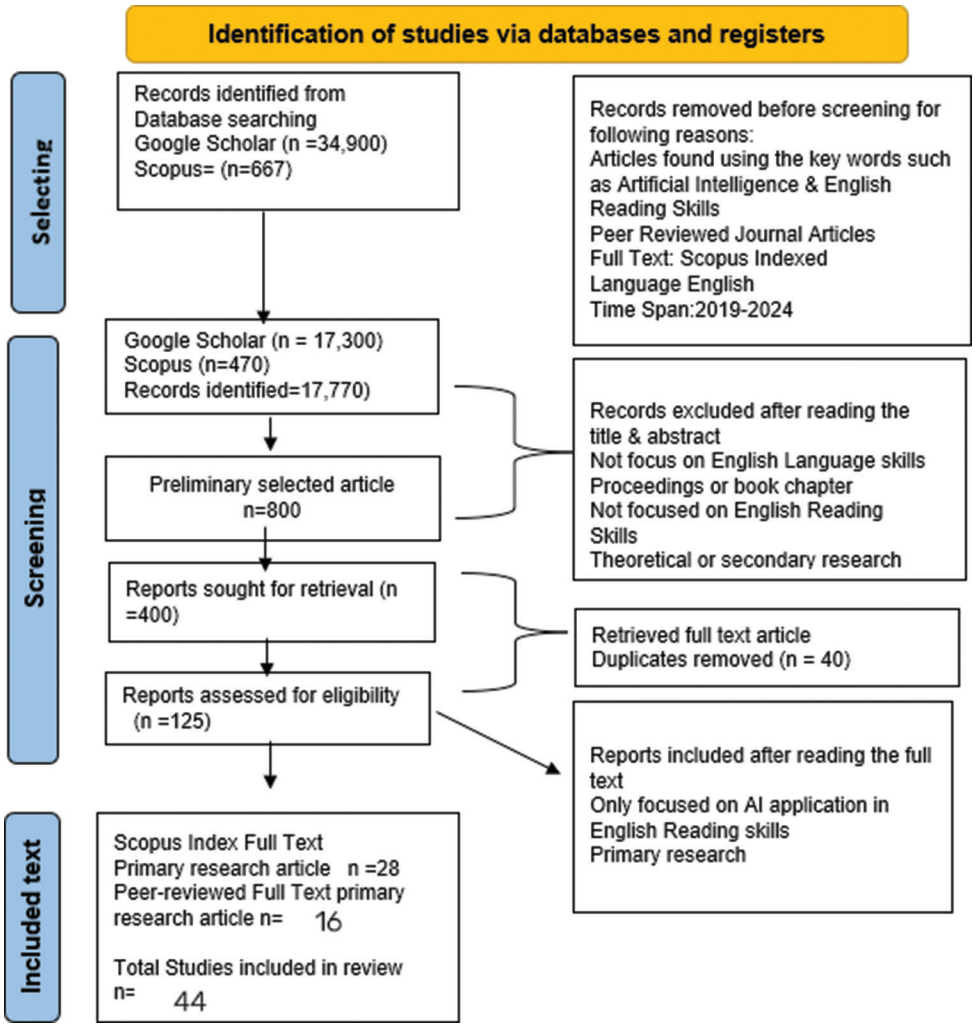


Figure 1. Flowchart of article review process (Page et al., 2021)

Data Coding and Analysis

The researchers classified the effectiveness of AIER in three main domains: Learning, Pedagogy and Assessment. Then, the theme emerging from the articles were synthesized in a Google Doc to categorize the effectiveness of the reviewed articles according to the main three domains. The reviewers then sat together, discussed among themselves and, taking the printed copies of the Google Doc, manually color-coded the three domains with initials 'L', 'P' & 'A'. For example, if an article focused on the ease of teachers' in using the machine learning system, it is categorized as 'A' (assessment). It is to be noted that one article may be classified in more than one domain.

Reviewers then started the coding process in an inductive approach. Three authors discussed the principles of thematic analysis and planned a coding process following Braun and Clarke (2006). To address RQ2, authors focused on the outcomes of using AI in reading in three main domains. Three authors read 44 primary research papers each and screened the full text. Each of them did the coding independently and extracted the sub-themes. While doing the meta-analysis, reviewers also noted the AI tool/technique that the researchers used in their studies. Later, all three authors had a meeting to discuss and confirm the sub themes. Finally, they summarized the categories and matched the domains, subthemes and technology technique. To retrieve the data for RQ1 and RQ3 the full papers were read focusing on the research theme & technology adoption and the contributions of the studies. All the authors made all the decisions consensually.

FINDINGS

Outcomes of Using AI in English Reading Pedagogy

Ten major roles of Using AI in English Reading (AIER) Pedagogy were identified from the reviewed articles across the three key domains: (i) Learning, (ii) Pedagogy, and (iii) Assessment. The contribution of the reviewed studies to each of these domains are described below.

AIER in student learning

The application of AI to student learning can be categorized into six main roles: (i) enhancing reading comprehension and vocabulary (ii) increasing learners' interest in motivation and reading (iii) Enjoyable and engaging learning process (iv) improved interactivity (v) Create Personalized Learning Environment and (vi) Multimodal learning experience.

1. Enhancing reading comprehension and vocabulary: AI-based environments have been shown to improve reading comprehension and vocabulary skills in language learners. Huang et al. (2023) found that the Intelligent Tutoring System (ITS) and Intelligent Tutoring Structure Strategy (ITSS) help students identify text structures and provide hints and thus help in comprehension. Hou (2021) found that the Mint Reading APP simulates a teacher role by testing students' vocabulary, comprehension, reading habits, and hobbies. Hsiao and Chang (2023) found that Taipei CooC-Cloud and AI tools like Linggle Read helped senior high school students select reading passages to improve their comprehension skills. Wang and Yan (2022) discovered that integrating a Production Oriented Approach (POA) improved college students' English reading levels. Studies show AI technology improves students' reading comprehension and vocabulary, but further research is needed to determine its effectiveness in classrooms worldwide.
2. Increase learners' interest and motivation in reading: AI technology can enhance learning motivation and interest in students. Studies show that Chinese school students' reading motivation improved after using AI robots. The Mint Reading APP improved vocational college students' reading comprehension skills. Liu et al. (2022) in their study found that using Chatbots in reading increased students' interest and social connection, leading to positive experiences. Personalizing learning materials with AI tools can enhance L2 proficiency and foster intrinsic motivation (Xiao & Hu, 2019).
3. Enjoyable and engaging learning process: Artificial intelligence (AI) can enhance the learning process by making it more engaging and pleasurable for learners. A study by Lee et al. (2023) found that AI-based content generators (AICG) can enhance the enjoyment of reading passages. Hsiao and Chang (2023) found that AI-powered tools can help students improve their reading and writing skills by allowing them to focus more on the lesson, avoid mind wandering, and engage more with their teachers and peers. Overall, AI tools can make learning more enjoyable and engaging for learners.
4. Improved interactivity: Artificial Intelligence (AI) tools have been used to record student learning data and enable interaction for a more flexible digital environment. For instance, a study by Liu et al. (2022) confirmed that Chatbot helped create a scope for positive experience and interaction for students. Another study by Alshriedeh (2021) concluded that persuasive reading with the help of Hot Bot strategy is crucial for learning second language accents and promoting literacy through interaction with content. Using Google Assistant for reading comprehension exercises also proved fruitful in promoting interaction as it incorporated fun and pleasure into learning (Sing et al., 2019). These results show that AI technologies help learners achieve better performance in reading skills because of improved interactivity with the materials.
5. Personalized Learning Environment: Recently, AI has garnered attention for providing students with personalized environments for an improved learning experience. The study done by Pettela (2020) and Srinivasan and Murthy (2021) show that AI can improve secondary level students' learning by providing them with personalized resources, such as presenting concepts related to reading passages, identifying letters and words through sound, and creating connections between texts, images, and sounds. The E-learning systems can also enhance primary school students' reading skills by providing personalized lessons (Sridama & Siribut, 2021) and personalized learning platform helped students to identify areas where they need additional support (Hidayat, 2024). The machine learning system and AI-based model have also been used to provide adapted personalized learning experiences for each student which enhances students' independent learning ability, increase reading ability and make the system more practical (Qianjing & Lin, 2021; Sridama & Siribut, 2021). The findings of Bunting & Segerstad (2020) showed that the teachers consider personalized learning technologies as a promising way to mitigate the competence differences between children in Sweden.
6. Multimodal learning experience**:** AI-powered tools are improving students' learning experiences compared to traditional methods. Research by Dong et al. (2022) explores multi-criteria decision support systems for AI-enabled English multimode online reading production and application. AI can enhance students' English proficiency and assist in updating their methods, enhancing their overall English learning experience. Srinivasan and Murthy (2021) suggest that Intelligent Tutoring Systems can enhance students' reading comprehension by providing a multisensory learning experience.

AIER in pedagogy

The application of AI can be divided into two main categories: (i) Leverage on teachers' role (i) Generate customized learning resources.

1. Leverage on Teachers' role**:** In the present pedagogical context, Artificial Intelligence is well-received not only by the learners but also by the instructors themselves. Studies show teachers having positive attitudes towards incorporating AI technologies in their classrooms (Bunting & Segerstad, 2020; Sumakul, Hamied & Sukyadi, 2022; Feng & Wang, 2023). Software such as Quillbot are being harnessed by teachers to enhance students' grammatical and analytical skills (Yudhiantara & Sugilar, 2023). Research by Feng and Wang (2023) shows that educational robots aid in bilingual reading proficiency, gaining positive feedback from parents, instructors, and students. This allows teachers to focus on improving teaching efficiency and student outcomes, thereby enhancing the effectiveness of AI in education.
2. Customized Learning Resources: Online intelligent English teaching platform, machine reading comprehension systems are used by the researchers in teaching reading comprehension offer learners with a wide range of relevant reading content. The study by Xiao, Y.,

Hu, J. (2019) suggests that the machine learning pedagogy directly impacts ESL reading by assigning materials based on students' interest linking new content to students' prior knowledge. By generating authentic reading materials, AI-based generative tools known as Copy AI & content generator (AICG) are reducing the workload of teachers (Lee, Shin & Noh, 2023). AI-based tools such CoAsker for question generating, AICG for reading content generation have also proved to have a positive impact on students' engagement with reading. Hence, Artificial intelligence has the capacity to provide a wide range of original English texts customizing as per the needs of the students that significantly can help English teachers.

AIER in assessment

The application of AI can be divided into two main categories: (i) Instant and Easily Available Feedback and (ii) Tailored Feedback.

1. Instant and Easily Available Feedback: In recent years, neural network based autonomous reading assessment, natural language processing (automatic speech recognition system) were found to be used in assessing reading performance and proved to be almost similar to human expert raters (Chen2024; Yıldız, et.al., 2024; Farangi, &, Zabbah, 2023). AI based tools such as CoAsker, Audience Response Systems (ARS) and Intelligent Tutoring System (ITS) are also being used to provide students with automated corrective feedback on their reading ability (Hu, 2023). For example, Pérez-Segura et al. (2022) found that the students using an AI powered tool called Clickers received personalized feedback that improved their receptive reading skills. In another study, AI-enabled speech recognition assessments demonstrated significant reading achievement and improvement in readers' engagement with instant feedback with an under-resourced education system (Bachiri et al., 2024). Feng and Wang (2023) in their research findings found positive outcomes in providing guidance and feedback while using AI-robot. These results illustrate how AI powered tools not only foster the growth of reading skills but also support students' learning by providing corrective feedback.
2. Tailored Feedback**:** Xu et al. (2019) in their review of 19 studies showed that Intelligent Tutoring Systems (ITS) can be as effective as humans. ITS can provide students with customized feedback that helps students to improve their reading comprehension skills. AI-based tools such as Audience Response Systems (ARS) have also been found to have beneficial impact in improving English skills providing customized feedback (Pérez-Segura et al., 2022). Rusmiyanto et al. (2023) in their research showed that students' communication skills were improved through customized feedback. Therefore, AI can be of great help in providing tailored feedback according to the needs of the students as well as in improving their reading comprehension.

Overall, the review analysis shows that AI is utilized in reading pedagogy that proved to have effective outcomes in terms of learning, materials, and assessments. Instead of being threatened by AI, proper utilization would eventually facilitate the teachers and the teaching process. The following chart shows the AI technology being adopted in different domains of teaching reading and its effectiveness from 2019-2024 (Figure 2).

DISCUSSION

The findings from the reviewed studies reveal that Artificial Intelligence (AI) plays a transformative role in English reading pedagogy enhancing English language learning outcomes (Jawaid et al., 2025). Over the past few years, particularly since 2021, AI has gained momentum in language education due to its multifaceted benefits across three core domains—student learning, pedagogy, and assessment. The findings also indicate that AI will increasingly be adopted in classrooms around the world to support reading comprehension, vocabulary development, sentence structure awareness, and learners' analytical abilities. In doing so, it will not only redefine traditional approaches to reading instruction, but also significantly will shape teaching practices and learning which is reinforced in the study of Son et al. (2025). Consequently, AI will become an integral part of language education processes.

The analysis identifies that one of the significant contributions of AI lies in its ability to personalize the learning experience. AI Tools provide a multimodal experience—combining text, audio, and visuals—to help students better engage with reading materials, making the overall learning process more effective and inclusive that can be a paradigm shift in teaching English reading skills (Jawaid et al., 2025). AI excels by delivering timely and personalized feedback; and can detect errors, provide corrective suggestions, and assess reading performance with an accuracy that approaches human-level evaluation- leading towards a transformative role for language instructors (Rohmiyati, 2025). AI also supports autonomous and interactive learning, enabling students to control their pace, revisit difficult concepts, and deepen their understanding through rich multimedia experiences. Importantly, AI doesn't just assist learners—it plays a vital role in pedagogical support as well. It contributes to the customization of learning materials and complements teachers' efforts in the classroom. By generating relevant content, offering instant feedback, and managing routine tasks, AI makes the teaching process more efficient and dynamic, allowing educators to focus more on higher-order instruction and learner interaction. Therefore, it is imperative for the teachers to integrate AI technology (Zhou & Hou, 2025).

The review also presents that AI technologies hold great promise for revolutionizing English reading pedagogy by supporting learning, assisting educators, and improving assessment practices. The key to maximizing this potential lies in adopting a critical and context-sensitive approach—one that values human judgment and cultural awareness while leveraging the power of AI to enhance instruction (Son et al., 2025). Language educators need to take the responsibilities

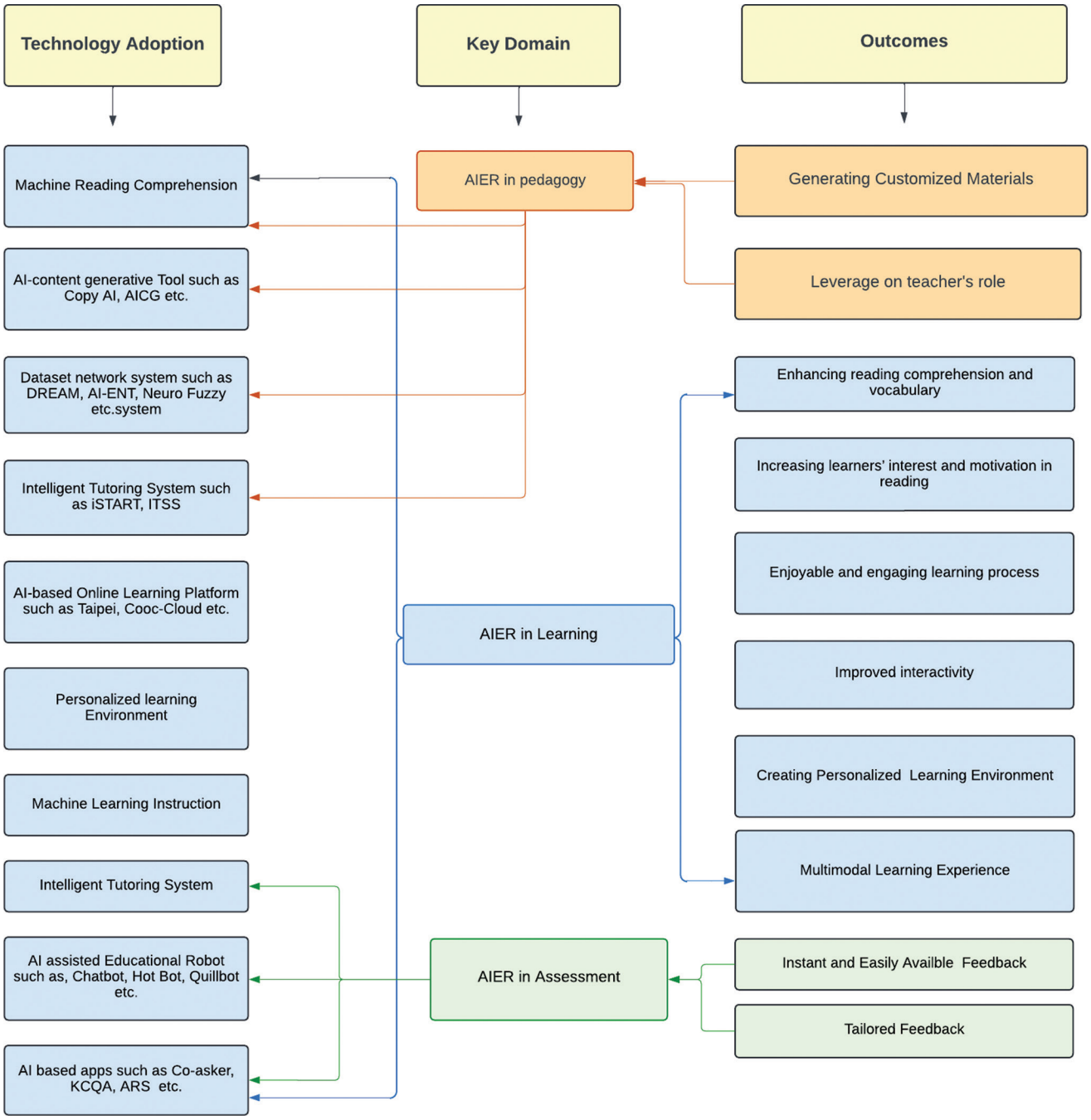


Figure 2. Outcomes of using aier along with the adopted technology

to ensure a balanced view in AI-driven environments that require acknowledging the challenges and limitations of AI integration. Rohmiyati (2025) reports that students are benefitting after using AI tools, apps, software; however, there are challenges and limitations in using AI such as understanding cultural nuances, unequal accessibility in all regions, risks of privacy and data security, excessive reliance on technology rather than interpersonal communication. While results are encouraging, there remains a need for further research to assess the broader applicability and effectiveness of these tools in diverse educational contexts around the world. With thoughtful implementation and ongoing research, AI can serve as a powerful ally in building more inclusive, effective, and future-ready reading classrooms.

FUTURE SCOPE

Using artificial intelligence to develop reading skills has shown positive outcomes. However, existing research shows some gaps which need to be addressed to fully integrate AI in classrooms. While many reviewed studies report short-term improvements in reading performance, future research should move beyond tool-based effectiveness and examine pedagogical sustainability over time, particularly through longitudinal designs. First, most current research focuses on tools instead of strong teaching-learning methods, and the evidence-based studies focus on short term results without looking at longitudinal studies to determine whether the benefits are truly sustainable. In this regard, future studies should

investigate whether prolonged exposure to AI-supported reading environments may lead to attention fragmentation, superficial reading, or reduced learner autonomy when reliance on AI becomes overly dominant.

Second, there is also a gap of empirical evidence on whether AI can develop higher order abilities like critical reading or inferencing despite claims about its potential to personalize learning and boost engagement. Moreover, while automated feedback could be beneficial to a certain extent, it may overlook the reading comprehension's intricate and human centered character. At the same time, heavy dependence on automated feedback may limit learners' chances to reflect and negotiate meaning through discussion, processes that are usually shaped through interaction with teachers.

Third, another significant concern is that there is an unequal distribution of the benefits of AI. Most research has been conducted on countries that have advanced technological infrastructure and which are in a better position to incorporate Artificial Intelligence tools into their education system, whereas low- and middle-income countries such as Bangladesh often do not have resources or infrastructural facilities to use these tools widely. Future research should therefore examine contextual constraints in low-resource settings, including limited access to devices, unstable internet connectivity, and teachers' varying levels of digital competence, to avoid reinforcing existing educational inequalities. Failure to address this technological gap may create a larger educational divide and limit how AI can be used broadly around the world. Existing research also fails to point out equity and accessibility issues at the student level. It is improbable that every student has access to computers, fast internet, or the skills to use AI software. By ignoring these factors, the current research shows a view of learning that is too reliant on technology which could make educational inequalities worse.

Another important concern is how AI reshapes teachers' professional roles and mental workload. While these tools are often advertised as saving time, the reality may present a different scenario. Monitoring AI tools, making sense of automated data, and adjusting AI generated materials to fit curriculum goals can quietly add to teachers' cognitive burden. There is also a lingering concern that teachers may have fewer chances to exercise and develop their own professional judgment, raising the possibility of erosion of their skills.

Finally, issues of ethical use of AI are also not addressed in the existing research. Future research should focus on creating AI tools that are based on practical teaching learning theories, sensitive to different contexts, and inclusive. It should also find ways to make automated feedback more personal, include lessons on ethics and privacy in schools, and think about AI in wider education policies. The limitations of AI in understanding culturally nuanced or open-ended tasks must be addressed. There is also a risk of over-reliance on AI, which could diminish the human interaction vital for developing critical thinking, empathy, and interpretive reading skills. With careful planning and proper training for teachers and students, AI can improve reading comprehension and make classrooms engaging and more focused on learners.

CONCLUSION

This systematic review examined recent studies on the use of artificial intelligence in English reading instruction, focusing on learner-centered applications, pedagogical practices, and assessment mechanisms. The review viewed AI integration as a mediational aid inside current reading pedagogy rather than as a substitute for instruction. Results indicate that AI methods are frequently employed to improve formative feedback, customization, and engagement in reading activities across different areas. However, because the three domains function interdependently rather than separately, their efficacy depends on teacher interpretation and well-informed instructional design. The review also highlights short-term, tool-focused data and inadequate consideration of learner dependency, instructor cognitive burden, and ethical issues. In order to investigate how AI may support human instruction in creating sustainable and context-sensitive English reading pedagogy, future research should use theory-driven and longitudinal methodologies.

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APPENDIX A. REVIEWED RESEARCH ARTICLES

Appendix A. Reviewed research articles

No	Authors (Year)	Article title	Journal name/ H-Index	Research participants	Methodology	Technology adoption	Findings
1	Yıldız et al. (2024)	Can Artificial Intelligence Identify Reading Fluency and Level? Comparison of Human and Machine Performance	*Reading & Writing Quarterly H-index 43*	120 fourth-grade students from public schools in Turkey	Quantitative	Artificial intelligence-based automatic speech recognition system	The findings of the study demonstrated that an automatic voice recognition system based on artificial intelligence can grade readers' fluency and accuracy as either good or poor, nearly matching the ratings of human experts. This study's findings open up a new way to evaluate oral reading abilities.
2	Chen (2024)	The Role of Artificial Intelligence in English Language and Literature Reading Management	*International Journal of Information and Communication Technology Education (IJICTE) H-index-21*	Chinese context	Quantitative	A user-autonomous evaluation system is created using artificial neural networks.	The study's findings indicate that an autonomous reading assessment system for English Language and Literature based on neural networks enhanced participants' reading efficiency by 0.48% and raised the standard of reading management for these subjects. The result also demonstrates that this approach effectively raises the standard of reading source sharing, lessens users' reading load, and improves their capacity for independent reading.
3	Hidayata (2024)	Effectiveness of AI-Based Personalised Reading Platforms in Enhancing Reading Comprehension	*Journal of Learning for Development H-index-9*	43 senior high school students in Indonesia	Quantitative	AI-based personalized reading platforms	The study's findings support the idea that improving reading comprehension can be achieved with an AI-based tailored reading platform. Improved learning outcomes can result from the adoption of AI-based learning systems which can provide students individualized learning experiences, pinpoint areas in which they require more assistance, and give real time feedback.
4	Rahman & Nor (2024)	Unveiling The Role of Artificial Intelligence on Reading Process	*Journal of English Education and Literacy*	Indonesian context	Qualitative	Three AI-based tools-Quillbot, Paraphraser.io, ChatGPT, Storywiz.io	Tools powered by artificial intelligence can help students with their reading. You can utilize ChatGPT for pre-reading exercises. For reading exercises, Quillbot and Paraphraser.io can be utilised. Post-reading activities can be conducted with Quillbot and Storywiz.io.

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Appendix A. *(Continued)*

No	Authors (Year)	Article title	Journal name/ H-Index	Research participants	Methodology	Technology adoption	Findings
5	Bachiri et al. (2024)	Integrating AI-based Speech Recognition Technology to Enhance Reading Assessments within Morocco's TaRL Program.	*Turkish Online Journal of Distance Education H-index-32*	100 Moroccan first to third graders	Mixed-method approach	Artificial intelligence-powered speech recognition technology	According to the study, assessments using AI-enabled speech recognition showed improvements in readers' engagement with immediate feedback in a under-resourced educational system as well as in basic literary development and reading achievement.
6	Jarrah (2024)	The reality of using the Artificial Intelligence-based reading progress tool in Microsoft Teams to enhance the reading skills among English language teachers of the Education Department in Jazan Region and the Obstacles to its use.	*Educational Sciences Journal*	146 English Language teachers from Saudi Arab	Quantitative	AI-based Reading Improvement Tool in Microsoft Team	According to the study's findings, teachers in the Jazan areas have a favorable opinion of the AI tool that Microsoft Teams offers since it adds resources to classroom instruction to help students' reading skills. However, in order to maximize the efficiency of the Microsoft application and make better use of the resources provided to suit kids' reading development, teachers require assistance.
7	Lee et al. (2023)	Artificial Intelligence-Based Content Generator Technology for Young English-as-a-Foreign-Language Learners' Reading Enjoyment	RELC Journal Sage Publication H-Index-42	five and six grade students (aged 11–12) learning English-as-a-foreign-language in South Korea		AI-based generative tool known as Copy AI & content generator (AICG) based activity were given	The results of the study demonstrate that the experimental group had a greater enjoyment of the reading passage learning process than the control group. In the long run, it will benefit the young students' development of information literacy and their ability to read extensively. By offering a large selection of original English texts that might be utilized in reading courses and for assessment, the AICG is anticipated to greatly lessen the workload of English teachers.
8	Liu et al. (2023)	Student-AI Question Co-Creation for Enhancing Reading Comprehension	IEEE Transactions on Learning Technologies H-index-58	59 first-year undergraduate students age from 19 to 21 years enrolled in a Learning Analytics class from China.	Quantitative	CoAsker question generation app & Human Computer Interaction (HCI) were used..	The results of the study show that students using this tool (treatment) were more engaged in generating low-level cognitive questions and performed better in acquiring knowledge than those using a traditional online question generation tool (control). These results indicate that student-AI question co-creation is beneficial to Student Question Generation training and educational assessment for reading comprehension, such as repeated practices.

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Appendix A. *(Continued)*

No	Authors (Year)	Article title	Journal name/ H-Index	Research participants	Methodology	Technology adoption	Findings
9	Huang et al. (2023)	Trends, Research Issues and Applications of Artificial Intelligence in Language Education	Educational Technology & Society H-index-103	Hong Kong	Quantitative	Intelligent tutoring systems (ITS) was used which used Apps such as Interactive Strategy Training for Active Reading and Thinking (iSTART), & Improve Teach Structure Strategy (ITSS).	The results showed that there were more studies on AI-enhanced language instruction over the period. The findings also showed that AI was widely utilized to help children acquire vocabulary, grammar, speaking, listening, writing, and reading. Learner profiling, automated speech recognition, and natural language processing were frequently used to create intelligent tutoring, tailored learning, and automated writing evaluation systems.
10	Hsiao & Chang (2023)	Enhancing EFL reading and writing through AI powered tools: design, implementation, and evaluation of an online course	Interactive Learning Environments H-index-57	43 students from 10 senior high schools in Taiwan.	Mixed method	Online platform called Taipei CooC-Cloud.& AI tools: Linggle Write, Linggle Read, Linggle Search.	According to the study, language competency predicted semester grade and assignment number, and students had a better overall experience in student-centered presentations than in teacher-centered lectures. As an illustration of how to create a high-quality synchronous online course on AI tools for EFL reading and writing using theoretical techniques, this study has both theoretical and practical relevance.
11	Feng & Wang (2023)	A comparative study on the development of Chinese and English abilities of Chinese primary school students through two bilingual reading modes: human-AI robot interaction and paper books	*Frontiers in Psychology* H-Index184	85 5th grade Chinese students	Quantitative	AI assisted educational Robot	The findings showed that, in comparison to their counterparts in the paper book class, students in the AI class showed notable improvements in bilingual reading motivation, reading quantity, reading comprehension, autonomous learning capacity, pronunciation competency, and test scores. Teachers, parents, and kids all expressed satisfaction with the AI-assisted reading mode that included instructional robots. In addition to allowing for the simultaneous development of multilingual fluency in both Chinese and English, it has the ability to successfully replace parental involvement in parent-child reading and English tutoring.
12	Liu & Huang (2023)	Retracted: Design of Artificial Intelligence-Based English Network Teaching (AI-ENT) System	Mathematical Problems in Engineering Hindawi H-index-78	School students	Quantitative	the artificial intelligence-based English network teaching (AI-ENT)	The framework of the study entails integrating the education into the classroom, offering support, and personally benefiting from it. According to the findings, students who participate in online forums improve their eloquence and acquire communication skills that they may apply outside of the classroom. Additionally, the proposed AI-ENT approach is more effective in assessing their English proficiency. By using this method, students can pinpoint and concentrate on their areas for growth. Students are more eager to learn when they are motivated and confident.

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No	Authors (Year)	Article title	Journal name/ H-Index	Research participants	Methodology	Technology adoption	Findings
13	Peng et al. (2023)	Examining ICT attitudes, use and support in blended learning settings for students' reading performance: Approaches of artificial intelligence and multilevel model	*Computers & Education H-index-17*	32 Organization for Economic Co-operation and Development countries from the latest cycle of the Programme for International Student Assessment (PISA) 2018 reading dataset	Quantitative	Combination of artificial intelligence algorithm and statistical analysis	The findings indicate that while school ICT use support for teachers and the teaching of ICT skills are negatively correlated with students' reading scores, students' perceived autonomy and ICT competence, preference for various reading formats, the quality of school ICT resources, and school ICT use policies are positively correlated with reading performance.
14	Bin-Hady et al. (2023)	Exploring the dimensions of ChatGPT in English language learning: a global perspective	*Library Hi Tech H-index 49*	Twenty ResearchGate researchers	Qualitative	RG Platform discussion feature	Students' language proficiency and systems can be improved with ChatGPT. The AIALL paradigm can support learning scaffolding, flexibility in the teacher's role, a fun learning environment, increased learner autonomy, future innovation encouragement, and celebration of diverse applications.
15	Rogers et al. (2023)	Qa dataset explosion: A taxonomy of NLP resources for question answering and reading comprehension.	ACM Computing Surveys H-index-213	200 datasets were categorized	qualitative	Descriptive analysis	The NLP community has created a large and rapidly growing collection of QA (question answering) and RC (reading comprehension) datasets. The most thorough survey of the field to date was published in this publication, which also identified the important dimensions. These factors offer a conceptual framework for assessing the format, domain, and target reasoning skills of both present and future resources. Both NLP practitioners searching for data and those trying to advance QA/RC research might find this survey helpful.
16	Farangi & Zabbah (2023)	Intelligent Scoring in an English Reading Comprehension Course Using Artificial Neural Networks and Neuro-fuzzy Systems	*TESL Quarterly*	Sixty-six students' scores in an English reading comprehension course were used for training the systems.	Quantitative.	Artificial Neural Networks and Neuro-fuzzy Systems	It was found that the NFS was able to predict and produce reading comprehension scores that aligned with the linear sum of students' scores. In addition, two-layer ANN had been outperformed by three-layer ANN.

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No	Authors (Year)	Article title	Journal name/ H-Index	Research participants	Methodology	Technology adoption	Findings
17	Sankumpa et al. (2023)	Using Constructivist Integrated Gamification and AI Assisted Learning Tool to Enhance Reading Comprehension for Fourth Grade Students	*Data Science and Institutional Research*	Participants included 6 and 9 pilot students	Quantitative	AI Chatbot	The findings demonstrated that the content was designed in accordance with reading comprehension principles and featured an AI chatbot to assist students with word meanings. The Collaboration Center promoted group work and encouraged the sharing of ideas to solve problems more effectively. The passing score for English reading comprehension was 70%.
18	Yudhiantara, & Sugilar (2023)	English Pre-Service Teachers' Ability to Use Artificial Intelligence-Based Application in Reading Comprehension	*Loquen: English Studies Journal*	Participants enrolled in the course "English for Islamic studies."	Qualitative	Quillbot	It was revealed that pre-service English teachers successfully used Quillbot to paraphrase articles which enabled improvement of students' reading comprehension. Students also used their analytical ability and examined Quillbot's grammar output.
19	Kostka & Toncelli (2023)	Exploring Applications of ChatGPT to English Language Teaching: Opportunities, Challenges and Recommendations	*Teaching English as a Second or Foreign Language*	Undergraduate students in public speaking graduate students in academic listening and speaking courses for international students in a pathways program	Quantitative	Implementation of ChatGPT in class activities	The result shows that there was a noticeable increase in students' participation in class discussions and peer engagement. Students appeared to be even more engaged and enthusiastic about the material. They appeared to take pleasure in analyzing ChatGPT output. While the majority of undergraduate and graduate students had positive opinions of ChatGPT, though there were some doubts and resistance to using the platform for educational purposes and questioned ChatGPT"s value for promoting learning and drew attention to its technological shortcomings.
20	Pérez-Segura et al. (2022)	The effect of personalized feedback on listening and reading skills in the learning of EFL	*Computer Assisted Language Learning*, H-index-63	Primary education Sixth grades (11-12 yrs old)	Quantitative	Audience Response System (ARS) such as Clickers is used.	Clickers were used for motivation in both the experimental and control groups. Compared to the control group, those who received personalized feedback, demonstrated greater improvement in terms of receptive skills.
21	Liu et al. (2022)	An analysis of children' interaction with an AI Chatbot and its impact on their interest in reading	*Computers & Education* H-index-215	68 fifth-grade students (11- and 12-year-old) in an elementary school in Taiwan,	Qualitative	Chatbot	The study explored the benefits of using Chatbot for a positive interaction among students and sustaining their interest in reading The social affective cues made learning activities enjoyable and engaging. Their performance was also improved by the book talk cue feature.

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No	Authors (Year)	Article title	Journal name/ H-Index	Research participants	Methodology	Technology adoption	Findings
22	Wang & Yan (2022)	College English Reading Teaching Integrating Production Oriented Approach from the Perspective of Artificial Intelligence	*Computation Intelligence and Neuroscience* H-index-70	College English students	Quantitative	Backpropagation Neural Network (BPNN) : an AI powered method to assess English reading level	The results differ more in Class A's performance before and after the experiment compared to class B. This demonstrates the effectiveness of incorporating POA in teaching methods for students with higher English proficiency. Compared to traditional methods the POA integrated teaching method proves more effective in improving college students' reading skills.
23	Dong et al. (2022)	AI based production and application of English multimode online reading using multi-criteria decision support system	*Soft Computing* H-index-102	middle schools' English teaching classes	Qualitative	Multi-criteria decision support systems for AI-enabled production and application	AI-based English multimode online reading platform can enhance learners' English skills and help them update their approaches, thereby enhancing their overall English education experience.
24	Sumakul et al. (2022)	Artificial Intelligence in EFL classrooms: Friend or Foe?	*LEARN Journal: Language Education and Acquisition Research Network H-index-10*	four EFL teachers in a university in Indonesia	Qualitative	AI apps	Teachers' perception of using AI in the classroom was positive. They believed that integration of AI in their teaching practice can help students become more proficient in English.
25	Wang (2022)	Computer-assisted EFL writing and evaluations based on artificial intelligence: a case from a college reading and writing course	Library Hi Tech *H-index-49*	188 students from China Agricultural University	Quantitative	AI powered deep learning analysis	It was found in the study that computer scoring feedback was more effective than teacher feedback. Moreover, by the end of the writing project, students showed greater improvement in both independent learning and writing skills.
26	Tang & Foley (2022)	A Case Study on the Effectiveness of Applying Content and Language Integrated Learning in an Artificial Intelligence English Reading Course	*Arab World English Journal*	45 undergraduate students at Chongqing Technology and Business University, China	Mixed method	AI English Reading Course	It was found that the AI English Reading Course was very much effective. After taking the course, students showed improvement in answering questions about AI development, products, and basic language-related AI concepts. This was reflected in their performance on "Multiple choice question' and 'Noun explanations'.

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Appendix A. *(Continued)*

No	Authors (Year)	Article title	Journal name/ H-Index	Research participants	Methodology	Technology adoption	Findings
27	Kim et al. (2022)	AI Student: A Machine Reading Comprehension System for the Korean College Scholastic Ability Test	*Mathematics*	285 passages from CSAT exam and 30 test students preparing for the CSAT	quantitative using data augmentation strategy	A KCQA model called AI student	The KCQA model was able to evaluate whether the questions from the reading passage were appropriate. In addition, the data augmentation process showed potential to enhance test takers' reading comprehension.
28	Hou (2021)	Research on Adopting Artificial Intelligence Technology to Improve Effectiveness of Vocational College English Learning	Journal of physics conference series H-index-99	Vocational college students from china	Quantitative	Mint Reading APP Was developed to play the role of an instructor.	English learning will become smarter and more accessible. As natural language processing technology and deep learning algorithms advance, students will see a major improvement in their ability to learn English.
29	Srinivasan, & Murthy (2021)	Improving reading and comprehension in K-12: Evidence from a large-scale AI technology intervention in India	Computers and Education: Artificial Intelligence H-index-17	1 million children and 1500 hundred teachers across 8 states in India on a large scale.	Quantitative	Read to Me (RTM) is an automatically trained AI engine which worked as instructor	ReadToMe improves students enrolled in the RightToRead program's English reading and comprehension. Students who enrolled in ReadToMe classes (Treatment) scored higher compared to students who did not use this platform for English learning (Control).
30	Qianjing & Lin (2021)	An Artificial Intelligent based Construction and Application of English Multimodal Online Reading Model	Journal of Intelligent & Fuzzy Systems H-Index-82	Chinese participants	Quantitative	Developed AI-based English multimodal reading online mode platform	The use of A-I based models can improve students' English scores through personalized learning. That increases learners' ability to learn independently and enhances their interests in learning English.
31	Haggag (2021)	Implementing Artificial - Powered Tools in Enhancing TOEFL_ ITP Test Takers' Reading and Writing Skills	*Journal of the Faculty of Education, Benha*	21 participants from South Valley University, in Hurghada, Egypt.	Quasi-experimental design,	Artificial Intelligence based tools	AI powered tools enhanced both reading and writing of the students. The participants were satisfied in using AI apps such as Chatbot and Google Lens because of the instant feedback which positively impacted on the learners.
32	Sridama & Siribut (2021)	Developing Thai Reading Skills for Primary School Students by Tailoring them to each Learner Using Machine Learning Technique.	*Turkish Online Journal of Qualitative Inquiry*	350 primary level students from different schools of Bangkok	Quantitative	Developed an E-learning system.	The students spent time reading with the e-Learning system than spent time reading with Thai books. This research showed that The machine learning system can increase the reading skills of primary school students and could provide adapted personalized reading lessons for each student.

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Appendix A. *(Continued)*

No	Authors (Year)	Article title	Journal name/ H-Index	Research participants	Methodology	Technology adoption	Findings
33	Alshriedeh (2021)	The Effect of an Artificial Intelligence Program on EFL Students on Reading Skill	IlKogretim Online-Elementary Education Online	10 students aged 8-15 Ankara, Turkey	Quantitative	Hot Bot strategy	The study presents that reading skills is highly connected to students' linguistic information and with the use of AI students can improve their technological knowledge as well as obtain new information.
34	Li et al. (2021)	An Intelligent System to Improve Vocabulary and Reading Comprehension Using Eye-tracking & Artificial Intelligence	*Computer Science & Information Technology*	University students	Quantitative	AI-based application on the basis of eye-tracking device	The study shows that the use of the app was successful as it increased user's experience with the app and improved and enhanced user's vocabulary in reading comprehension.
35	Yin (2021)	Research on the Impacts of Artificial Intelligence Technology on Language Teaching Technology	*Frontiers in Educational Research*	982 students from Shanghai Publishing and Printing College	Quantitative	the smart English teaching platform	The study shows that AI-based online classes are more helpful for students in improving reading skills as it can offer autonomous learning resources, interesting and interactive teaching contents and methods relevant to the students' background and study major..
36	Lestari et al. (2021)	The Correlation between Students' Artificial Intelligence and Their English Reading Skills Achievement	*Bahasa: Journal Keilmuan Pendidikan Bahasa Dan Sastra Indonesia*	30 students in an English Education study Program	Quantitative	Mobile Apps	The study results show that a positive correlation exists between students' English reading achievements and students' use of AI in English education study program at Islamic University of Kalimantan Muhammad Arsyad Al Banjari Banjarmasin.
37	Bunting & Segerstad (2020)	Swedish Teachers' Views on the Use of Personalized Learning Technologies for Teaching Children Reading in the English Classroom	*International Journal of Child-Computer Interaction* H-index-33	Children in between 4-6 grade and 11 school teachers	Qualitative thematic analysis	AI-based system was developed	The result presents the experience of Swedish English teachers where the teachers demonstrated a noticeable difference in English competence of Swedish children after using AI. It also shows that AI-based personalized learning technologies can act as a promising technique to reduce the competence differences between children in Sweden.
38	Sun et al. (2020)	Design of Online Intelligent English Teaching Platform based on Artificial Intelligence Technique	*Computational Intelligence H-index-57*	School level students	Quantitative	Deep learning-assisted online intelligent English teaching system was developed	The findings demonstrate that the design helps students in improving English Language proficiency and can create learning content more relevant to the students.

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Appendix A. *(Continued)*

No	Authors (Year)	Article title	Journal name/ H-Index	Research participants	Methodology	Technology adoption	Findings
39	Xu et al. (2019)	The Effectiveness of Intelligent Tutoring Systems on K-12 Students' Reading Comprehension: A meta-analysis	*British Journal of Educational technology* H-index-110	10,000 K 12 students.	Quantitative analysis	6ITS programs were used	AI can work for Tutoring the students in improving the reading comprehension skills. ITSs can offer a practical choice for delivering dependable and easily accessible reading comprehension training.
40	Xiao & Hu (2019)	Assessment of Optimal Pedagogical Factors for Canadian ESL Learners' Reading Literacy through Artificial Intelligence Algorithms	*International Journal of English Linguistics* H-index- 45	203 participants from 128 primary schools	Quantitative analysis	*Machine learning me*thod SVM	ESL reading teaching can improve primary school readers by assigning materials based on students' interests. The study suggests that the machine learning pedagogy of in-class activities directly impacts on the effectiveness of ESL reading teaching. This improves reading comprehension and facilitates memorizing. teachers may be able to apply diverse pedagogies in large classes.
41	Bin & Mandal (2019)	English Teaching Practice based on Artificial Intelligence Technology.	*Journal of Intelligent & Fuzzy Systems* H-index-85	Middle school students	Qualitative	AI-based English teaching instruction system was designed	The study presents that teaching English with AI-based instructions creates a better opportunity for the optimization of the quality and effect of the English teaching process and for personalized learning.
42	Sing et al. (2019)	Ask the Assistant: Using Google Assistant in Classroom Reading Comprehension Activities	*International Journal of New Technology and Research*	Ten-year-old participants from a school in Malaysia	Qualitative analysis	Google Assistant	The effectiveness of Google Assistant has been demonstrated to some extent by three measurable indicators: the quantity of questions asked, the accuracy of the answers provided, and the amount of time required to finish comprehension tasks. There have been significant improvements in participants" perceptions of the reading comprehension as a result of using Google Assistant. Users find it more pleasurable, fun and interactive.
43	Abba et al. (2019)	Use of Computer Assisted Instruction to Improve Students' Reading Skill in English Language	*I-manager's Journal on English Language Teaching*	40 students from Nigeria	Quasi-experimental design	Computer Assisted Instruction	The findings showed that computer-assisted instruction positively helped students to develop learners' comprehension skills in reading. The results show that students in the experiment group were better at remembering vocabulary and had better reading comprehension skills than the control group due to the use of computer assisted instruction.
44	Sun et al. (2019)	DREAM: A Challenge Data Set and Models for Dialogue-based Reading Comprehension	*Transactions of the Association for Computational Linguistics H-Index-56*	Variety of English Language Exams were used	Quantitative	DREAM, a dialogue based dataset was developed	DREAM enables learners to comprehend a few structural patterns of dialogue better. It facilitates reading comprehension by including general world knowledge and characteristics of the dialogue format. It combines tasks like matching and reasoning.